CLEANCITY
AI-POWERED WASTE MANAGEMENT ECOSYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment for the requirements for the award degree of

Bachelor of Engineering

CSB1332 - DESIGN PROJECT

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K RAMAKRISHNAN COLLEGE OF TECHNOLOGY (AUTONOMOUS)

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MAY 2026

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**CLEANCITY: AI-POWERED WASTE MANAGEMENT ECOSYSTEM**" has been carried out by the students, **SANTOSH G V, SHERWIN B, SIVA SURYA S, SRI HARISH.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce: 5.5.2026

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ABSTRACT

CleanCity is an innovative, AI-driven web ecosystem designed to modernize urban waste management by seamlessly connecting citizens, municipal authorities, and waste collectors. Traditional waste disposal suffers from source-level contamination and inefficient, static collection routes that inflate operational costs. To address these challenges, CleanCity integrates edge-based artificial intelligence, utilizing a custom Google Teachable Machine model deployed via TensorFlow.js. This allows citizens to instantly scan and classify household refuse into Solid, Bio-waste, or E-waste directly through their mobile browser without server latency. The system features a centralized scheduling platform where users can geo-tag pickup requests, earning "Green Points" through a gamified reward system upon verified collection. For municipal authorities, the platform provides a data-driven command center featuring live request heatmaps and integrates the Open Source Routing Machine (OSRM) API to dynamically generate mathematically optimized, multi-stop driving routes for collection fleets. Furthermore, a dedicated mobile interface for waste collectors ensures operational transparency by mandating photographic proof of task completion. By replacing outdated manual methods with an integrated digital workflow, CleanCity eliminates source contamination, drastically reduces municipal fuel consumption, and fosters a sustainable, community-driven economy.

Keywords: AI Waste Classification, Dynamic Route Optimization, Smart City Sanitation, Edge Computing, Behavioral Gamification, OSRM Pathfinding, Circular Economy.

ACKNOWLEDGEMENT

We thank our **Dr. N. Vasudevan**, Principal, for his valuable suggestions and support during the course of my research work.

We thank **Mr. R. Rajavarman**, Head of the Department, Assistant Professor (Sr. Grade), Department of Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Ms. P. Karthika**, Assistant Professor, Department of Computer Science and Engineering for her keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mrs. S. Gayathri**, Assistant Professor, Department of Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Department of CSE, K Ramakrishnan College of Technology(Autonomous), Trichy, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
B2C	-	Business-to-Consumer
API	-	Application Programming Interface
B2G	-	Business-to-Government
CPU	-	Central Processing Unit
CSS	-	Cascading Style Sheets
CV	-	Computer Vision
DOM	-	Document Object Model
ETA	-	Estimated Time of Arrival
GPS	-	Global Positioning System
HTML	-	Hypertext Markup Language
IoT	-	Internet of Things
JSON	-	JavaScript Object Notation
ML	-	Machine Learning
OSRM	-	Open Source Routing Machine
RBAC	-	Role-Based Access Control
UI	-	User Interface
UX	-	User Experience
VRP	-	Vehicle Routing Problem

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

CleanCity is a comprehensive, AI-powered waste management web application engineered to modernize urban sanitation by seamlessly connecting citizens, municipal authorities, and waste collectors. Operating on a unified digital ecosystem, the platform addresses the logistical bottlenecks of traditional waste disposal by introducing real-time data tracking, automated route optimization, and edge-based artificial intelligence for source-level segregation.

The system empowers everyday users through a dedicated Citizen Dashboard, transforming waste disposal into an interactive and rewarding process. The flagship feature of this interface is the AI Waste Scanner, which leverages a custom Google Teachable Machine model deployed via TensorFlow.js. This tool utilizes the user's device camera to instantly scan and classify loose refuse into Solid, Bio-waste, or E-waste categories directly within the browser. Citizens can then pinpoint their exact geographical location to schedule a pickup, track the real-time approach of their assigned collector on a live map, and consult a digital segregation guide for specific disposal protocols. To ensure sustained civic engagement, the platform employs a gamification framework that awards users "Green Points" for successfully completed pickups, which can be tracked and redeemed for rewards.

For city administrators, the framework provides a robust Authority Dashboard designed for macro-level logistical oversight. Authorities can actively monitor incoming pickup requests and assign them to available collectors. To visualize city-wide waste generation, the dashboard features a Live Waste Heatmap powered by Leaflet.js, which calculates request density and highlights critical extraction zones.

To complete the operational loop, the ecosystem includes a mobile-responsive Worker Dashboard tailored for waste collectors. This interface allows drivers to view their assigned jobs, initiate map-based navigation to target coordinates, and upload photographic evidence of the collected waste to formally close the task loop and trigger the citizen's point reward.

1.2 PROBLEM STATEMENT

Despite growing environmental awareness, urban waste management continues to be hindered by systemic inefficiencies at both the citizen and administrative levels. A primary challenge is the lack of accurate waste segregation at the source, as everyday citizens frequently struggle to correctly classify complex household refuse into specific categories such as Solid, Bio-waste, or E-waste. Without immediate, accessible guidance to assist in this categorization, contaminated waste streams are routinely produced, which severely disrupts downstream recycling processes and sustainability efforts.

Compounding this issue is the logistical disconnect experienced by municipal authorities. City administrators traditionally operate with a severe lack of real-time geospatial data regarding where and when waste is accumulating. As a result, waste collection fleets are deployed on static, unoptimized routes that fail to adapt to daily fluctuations in pickup demands. This reliance on outdated routing methods leads to excessive fuel consumption, increased operational latency, and the deployment of resources to areas with minimal waste density.

Furthermore, the operational chain connecting citizens, administrative dispatchers, and field collectors suffers from a critical lack of transparency and accountability. Citizens are often left without any mechanism to track the real-time status or arrival of their scheduled collection vehicles. Conversely, authorities and fleet managers lack streamlined tools to monitor active collectors or definitively verify the successful completion of assigned pickups through photographic proof.

Finally, there is a pervasive lack of civic motivation and behavioral incentivization. Without a tangible, rewarding framework to encourage eco-friendly habits, citizens possess little inherent motivation to actively participate in proper waste segregation or utilize formal pickup scheduling protocols. A comprehensive system is urgently required to bridge this gap by providing AI-driven segregation assistance, dynamic logistical optimization, and gamified civic rewards within a unified digital environment.

1.3 DOMAIN SPECIFICATION

The CleanCity project operates at the intersection of Environmental Sustainability, DeepTech, and Smart City Logistics. It leverages advanced Artificial Intelligence, specifically Machine Learning and Computer Vision, alongside geospatial mapping to modernize urban sanitation and foster a robust circular economy. The framework provides a dual-sided technological solution, bridging Business-to-Consumer (B2C) applications with Business-to-Government (B2G) data administration.

1.3.1 Edge AI and Computer Vision Domain

This domain focuses on eliminating the cognitive friction associated with household waste segregation through real-time image processing. The system utilizes lightweight object detection models deployed directly to the edge. By analyzing a live camera feed using a custom Google Teachable Machine model integrated via TensorFlow.js, the architecture instantly identifies and classifies household refuse into distinct categories such as solid, bio-waste, and e-waste.

1.3.2 Geospatial Routing and Smart Logistics Domain

This domain addresses the inefficiencies of municipal waste collection by transitioning fleets from static schedules to dynamic, data-driven pathfinding. It involves the aggregation of real-time citizen pickup coordinates into a centralized administrative dashboard. Utilizing spatial mapping libraries like Leaflet.js and the Open Source Routing Machine (OSRM) API, the system calculates the most mathematically efficient, multi-stop collection routes.

1.3.3 Behavioral Gamification and Civic Analytics Domain

This domain encompasses the psychological engagement of users and the macro-level analysis of civic data. To drive continuous user adoption and enforce proper disposal habits, the architecture incorporates gamification frameworks, rewarding verifiable eco-friendly actions with digital "Green Points". Simultaneously, the system structures complex, incoming civic data into interactive visual heatmaps and predictive analytics charts (via Chart.js).

1.4 AIM AND OBJECTIVE

1.4.1 Aim

The primary aim of the CleanCity project is to engineer an integrated, AI-driven software ecosystem designed to solve the critical challenges of urban waste segregation and municipal collection logistics. The platform strives to modernize municipal sanitation by bridging the gap between citizen disposal habits and administrative fleet management. By providing accessible edge-AI tools for the public and dynamic routing dashboards for authorities, the system aims to eliminate source-level contamination, establish a robust circular economy, and significantly reduce the carbon footprint and operational costs associated with traditional waste collection.

1.4.2 Objectives

The specific objectives of the CleanCity platform are:

- **Implement Edge-AI for Instant Segregation:** To develop and deploy an AI Waste Scanner using a Google Teachable Machine model (via TensorFlow.js) that allows citizens to instantly categorize household waste (Solid, Bio-waste, E-waste) using their mobile device cameras, entirely within the browser.
- **Enable Data-Driven Fleet Optimization:** To integrate the Open Source Routing Machine (OSRM) API to dynamically generate mathematically optimized, multi-stop collection routes for municipal garbage trucks based on aggregated, real-time citizen pickup coordinates.
- **Develop a B2G Civic Analytics Dashboard:** To construct a centralized administrative command center that visualizes incoming civic data through Live Waste Heatmaps (Leaflet.js) and statistical charts (Chart.js), empowering authorities to monitor neighborhood request densities and track collector completion rates.
- **Establish a Transparent Collector Workflow:** To provide an optimized mobile interface for waste collectors that enables turn-by-turn navigation to assigned locations and mandates the upload of photographic proof to verify task completion.

1.5 SCOPE OF THE PROJECT

The scope encompasses the development and deployment of a dual-sided, AI-integrated software ecosystem designed to modernize municipal waste management logistics and enhance public participation. The framework primarily targets urban households for source-level segregation and municipal authorities for backend fleet management, ensuring a seamless flow of actionable data from the initial point of disposal to the final collection route. By focusing on a web-based, software-first approach, the architecture eliminates the immediate need for expensive physical IoT hardware retrofitting in public bins, relying instead on the existing mobile device infrastructure of the general public.

A fundamental component of the framework involves the integration of edge-based artificial intelligence for real-time object classification. The scope specifically includes the curation and training of a custom computer vision dataset utilizing Google Teachable Machine. This methodology facilitates the rapid development of a lightweight classification model capable of identifying loose, localized household refuse. By leveraging the Teachable Machine platform for model training, the system ensures that complex image processing and inference occur directly within the user's browser via TensorFlow.js. This allows the application to instantly categorize scanned items into solid, biological, and electronic waste streams with high confidence scores, completely bypassing high server-side computational latency.

On the administrative spectrum, the scope extends to dynamic geographical tracking and logistical pathfinding. It covers the capability to ingest geolocated pickup requests and seamlessly convert them into mathematically optimized driving routes utilizing the Open Source Routing Machine (OSRM) API. This robust backend infrastructure enables city administrators to actively monitor fleet performance, track the status of assigned pickups, and evaluate neighborhood eco-compliance through an integrated green-point reward system. Finally, the scope includes a dedicated mobile interface for waste collectors to ensure accountability via photographic proof of task completion.

CHAPTER 2

LITERATURE SURVEY

2.1 A RELIABLE AND ROBUST DEEP LEARNING MODEL FOR EFFECTIVE RECYCLABLE WASTE CLASSIFICATION

The research work titled "A Reliable and Robust Deep Learning Model for Effective Recyclable Waste Classification" focuses on developing an automated waste sorting system capable of accurately categorizing distinct waste types at industrial facilities. Traditional manual sorting processes are highly inefficient and prone to human error, necessitating robust technological intervention at backend material recovery facilities. By applying deep learning to visual data, the system is designed to efficiently handle large volumes of refuse and categorize them into six distinct classes to streamline the recycling pipeline.

The system primarily relies on advanced computer vision techniques and deep learning architectures to process and analyze waste imagery. It evaluates models such as DenseNet201 and MobileNet-v2, ultimately proposing the RWC-Net model for enhanced feature extraction. Additionally, the architecture leverages saliency maps, specifically Score-CAM, to gain visual insights into how the models identify critical features within datasets like TrashNet.

A key contribution of this approach is its remarkable precision, achieving an accuracy rate of over 95% in categorizing recyclable materials. This high level of reliability ensures that industrial sorting machines can operate autonomously, drastically improving the throughput and financial viability of downstream recycling operations.

In conclusion, the paper demonstrates that deep learning-based visual classification significantly enhances backend recruitment efficiency by automating the screening of waste materials. While highly effective for industrial plants, the approach relies entirely on heavy machinery and centralized processing, highlighting the need for decentralized, mobile-edge solutions—such as those utilizing TensorFlow.js and Google Teachable Machine—that empower source-level segregation by everyday citizens prior to collection.

2.2 REAL-TIME WASTE DETECTION AND CLASSIFICATION USING YOLOV12-BASED DEEP LEARNING MODEL

The research work titled "Real-Time Waste Detection and Classification Using YOLOv12-Based Deep Learning Model" focuses intensely on providing a phenomenally fast, scalable, and real-time computational solution for autonomous waste identification deeply embedded within smart city infrastructures. As modern urban environments continually transition toward complete logistical automation, the demand for exceptionally fast inference speeds is hyper-critical. Systems must be capable of processing live, high-definition video feeds directly from municipal security cameras and rapidly moving autonomous sorting robots without dropping frames or experiencing latency.

To achieve this, the system evaluates and utilizes highly advanced machine vision frameworks. The researchers conduct a rigorous performance comparison of legacy object detection models, including the Single Shot Detector (SSD) and Faster R-CNN, measuring them against the state-of-the-art YOLO (You Only Look Once) architectural family. By ultimately implementing the highly advanced YOLOv12 framework, the architecture heavily capitalizes on novel anchor-free bounding boxes and sophisticated high-to-low resolution image processing. This allows the model to accurately detect, localize, and classify highly complex, overlapping waste items almost instantaneously in a live video stream.

In conclusion, the application of the YOLOv12-based deep learning model represents a massive, significant advancement in the field of autonomous municipal sanitation and real-time computer vision. However, while the speed is unmatched, the system inherently requires a tremendously heavy computational load. This hardware reliance makes it best suited strictly for high-powered embedded industrial systems or heavily computerized autonomous machines. This indicates a clear, unaddressed gap in the consumer market for extremely lightweight, edge-AI software applications that are capable of running natively and efficiently within standard consumer mobile web browsers entirely without server reliance or heavy GPU requirements.

2.3 IOT-BASED SMART WASTE MANAGEMENT SYSTEM: A SOLUTION FOR URBAN SUSTAINABILITY

The research work titled "IoT-Based Smart Waste Management System: A Solution for Urban Sustainability" focuses on radically optimizing the physical logistics of waste collection to directly and measurably reduce urban fuel consumption and greenhouse gas carbon emissions. Traditional municipal sanitation fleets globally operate on highly inefficient, static time-based schedules. In these outdated paradigms, heavy diesel trucks are dispatched to empty public bins regardless of their actual volumetric capacity, which directly leads to massive logistical inefficiencies, wasted operational budgets, and unnecessary vehicular pollution.

To combat this, the system introduces a comprehensive, physical Internet of Things (IoT) infrastructure layer across the city. This involves retrofitting thousands of public bins with specialized hardware sensors to conduct continuous, real-time fill-level monitoring. As bins fill up, the sensor data is transmitted to a central server and then processed through highly intelligent mathematical route optimization algorithms. By dynamically altering the driving paths based on live data, the system ensures that garbage trucks are only ever dispatched to service bins that have officially reached a critical, pre-defined capacity threshold.

The environmental and financial results of this approach are highly significant. The integration of this hardware-software ecosystem successfully cuts total municipal truck driving mileage by an impressive 20%. Consequently, by optimizing these travel distances and minimizing engine idling time, the system simultaneously achieves a massive reduction in total fleet CO2 emissions by 18%, thereby maximizing public resource allocation.

In conclusion, IoT-based smart waste management provides a highly effective, scientifically validated logistical solution for urban environmental sustainability. While the routing improvements and emission reductions are undeniably substantial, the architecture inherently relies heavily on extremely expensive physical hardware sensors, continuous battery maintenance, and large-scale, disruptive municipal retrofitting. This suggests a massive opportunity for alternative, software-centric routing solutions that are driven entirely by predictive, citizen-generated digital pickup requests rather than fragile physical hardware.

2.4 WASTE CLASSIFICATION USING YOLO ON RASPBERRY PI

The research work titled "Waste Classification using YOLO on Raspberry PI" focuses on leveraging the power of localized edge microcontrollers and computer vision to fully automate physical waste separation directly at the individual bin level. To successfully eliminate the persistent issue of human error in public disposal scenarios—such as in busy commercial office buildings, airports, or automated smart homes—technological automation must occur instantaneously at the exact moment a user attempts to discard an item.

To achieve this point-of-disposal automation, the system seamlessly employs ultralytics deep learning architectures, specifically designing tests to compare the efficiency and frame rates of YOLOv5, YOLOv7, and YOLOv8 models. These highly compressed object detection models are deployed directly onto a Raspberry Pi microcomputer, which is actively connected to a live web camera feed positioned over the bin's opening.

The physical sorting mechanism is entirely driven by the AI's real-time inference. Upon successfully identifying and categorizing an object via the camera feed, the system utilizes backend Python scripts to immediately send electrical signals to control automated servo motors. These motors then mechanically actuate internal flaps, automatically routing the dropped refuse into distinct, physically separated internal compartments explicitly designated for recyclable, non-recyclable, and hazardous materials. This localized automation provides immediate, flawless separation of waste streams entirely without requiring any prior segregation knowledge or deliberate sorting action from the user.

In conclusion, deploying YOLO algorithms directly onto microcontrollers offers an incredibly innovative, futuristic hardware solution for the development of smart automated bins. However, executing and deploying this mechanical system on a massive city-wide scale necessitates the exorbitant manufacturing, continuous maintenance, and physical deployment of highly specialized mechanical hardware. This severe scalability bottleneck points directly towards a parallel, urgent need for zero-hardware, purely software-based alternatives that utilize existing consumer smartphone cameras to provide the same point-of-disposal guidance

2.5 RESEARCH PAPER ON IOT-BASED SMART WASTE MANAGEMENT SYSTEM FOR SMART CITIES

The research work titled "Research Paper on IOT-based Smart Waste Management System for Smart Cities" aggressively addresses the massive, macro-level infrastructure planning required by municipal governments to fundamentally mitigate illegal open dumping and forcefully streamline highly complex collection pathways. Rather than focusing on single bins or individual users, this research looks at the city as a singular, interconnected living organism that requires vast networks of synchronized data.

A highly significant advantage of this macroscopic, city-wide approach is the extreme, unprecedented enhancement of civic transparency it provides to city administrators. By integrating cutting-edge concepts like digital twins (creating a live virtual replica of the city's sanitation network) and secure blockchain ledgers for immutable data logging, the system allows for the rapid, automated detection of environmental hazards through high-level AI-driven surveillance.

This holistic, macro data overview subsequently empowers city planners to confidently draft long-term, highly accurate, and mathematically sound sustainability policies.

A significant advantage of this macroscopic approach is the extreme enhancement of civic transparency and the rapid detection of environmental hazards through high-level AI-driven automation. The holistic data overview enables city planners to draft long-term, highly accurate sustainability policies.

In conclusion, aggressively integrating digital twins and blockchain technology into smart city waste management creates a virtually impenetrable, highly secure logistical overview for government bodies. However, this advanced framework natively represents a highly complex, monumentally capital-intensive infrastructure overhaul that can take years to deploy. This massive barrier to entry intensely highlights the strict necessity for highly accessible, rapidly deployable, and localized software solutions that can be utilized immediately by smaller or developing municipalities without massive initial funding.

2.6 A REVIEW ON REAL-TIME WASTE TRACKING AND ROUTE OPTIMIZATION USING CLOUD-BASED IOT

The research work titled "A review on real-time waste tracking and route optimization using cloud-based IoT" focuses intensely on academically solving highly complex logistical routing problems in municipal waste management through the strict application of multi-criteria decision support algorithms. Logistics dispatchers and large fleet management companies absolutely require backend systems that can seamlessly balance multiple dynamic, shifting variables simultaneously to continuously ensure maximum fleet operational efficiency.

The study deeply explores the integration of highly complex, heavy mathematical models and advanced metaheuristics designed specifically to adapt entire collection routes dynamically while trucks are already in transit. It rigorously evaluates the practical use of the TOPSIS method, the bio-inspired Bat Algorithm, and complex Genetic Algorithms. These algorithms are tasked to continuously recalculate driving paths based on rapidly shifting, real-world criteria, which crucially includes weighing material toxicity risks, updating volumetric fill levels of target bins, and adhering to strict municipal time constraints.

The practical application of these theoretical algorithms provides undeniably highly optimized fleet vehicle management protocols. By allowing computer systems to handle the routing logic rather than human dispatchers, the methodologies drastically reduce operational latency and minimize unnecessary fuel expenditure across the entire grid.

In conclusion, the review strongly proves that cloud-based multi-criteria mathematical algorithms deliver exceptional, mathematically proven routing efficiency for handling complex municipal fleets. However, the extreme computational reliance on heavy mathematical metaheuristics naturally paves the exact way for modern, streamlined, and lightweight API-based pathfinding solutions. By transitioning to pre-built logic infrastructures, such as utilizing the Open Source Routing Machine (OSRM) API, platforms can achieve the exact same logistical optimization without the heavy localized computational overhead.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

In the current civic landscape, municipal solid waste management relies heavily on outdated, manual segregation protocols and static logistical frameworks. Households and everyday citizens lack immediate, accessible digital tools to assist in the precise categorization of complex waste streams at the point of disposal. As a result, the default behavioral pattern frequently defaults to depositing unsegregated solid, biological, and electronic waste into single, mixed bins out of convenience or confusion. Existing technological interventions often fail to provide an educational or interactive layer to guarantee accurate usage by the public.

Furthermore, current municipal collection operations deploy garbage trucks on fixed, pre-determined routes regardless of the actual waste volume generated in a specific geographic zone. This analog, disconnected approach means that city authorities lack real-time visibility into neighborhood-level waste generation, preventing them from dynamically optimizing fleet deployments. When citizens attempt to find proper disposal methods or schedule pickups, they are forced to rely on fragmented, inefficient communication channels rather than a unified digital platform.

The absence of an integrated, data-driven ecosystem creates a massive disconnect between the point of waste generation and the endpoint of waste processing. Without a centralized digital platform to track disposal habits, offer instant visual classification, or aggregate localized pickup requests, municipalities are entirely blind to the ground-level realities of their sanitation networks. Consequently, downstream recycling facilities receive highly contaminated refuse streams, making material recovery economically unviable and perpetuating a linear, landfill-dependent waste cycle.

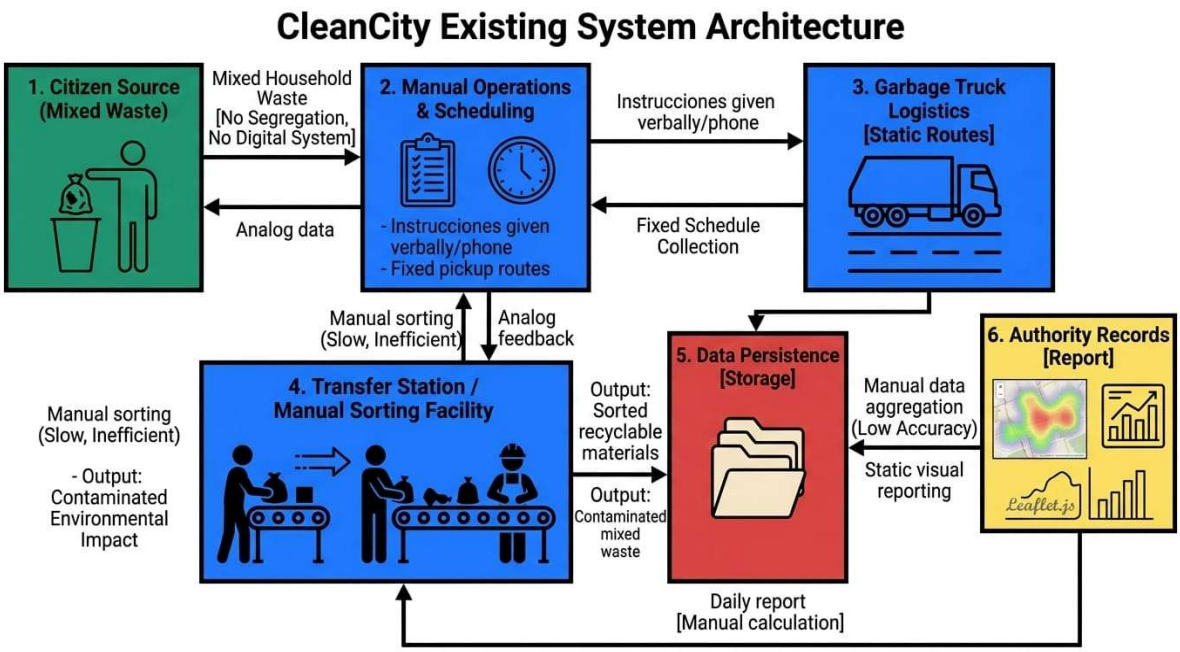

Figure 3.1 Existing System

3.1 DISADVANTAGES

- **Severe Contamination:** Recyclable materials are frequently ruined due to the improper mixing of wet bio-waste and hazardous electronic waste at the household level.
- **Logistical Inefficiency:** Municipal authorities face escalated operational and fuel costs stemming from static collection routes that ignore real-time waste density.
- **Cognitive Burden:** There is a complete absence of real-time, point-of-source guidance or AI-driven identification tools to help citizens navigate complex segregation rules.
- **Administrative Blind Spots:** The lack of a centralized, data-driven Business-to-Government (B2G) dashboard prevents city planners from utilizing actionable insights, analytics, and interactive heatmaps.
- **Zero Incentivization:** Current systems offer no gamification or economic rewards to motivate households to consistently participate in proper waste disposal practices.

CHAPTER 4

PROBLEMS IDENTIFIED

Based on a thorough analysis of the existing waste management infrastructure and logistical frameworks, several critical systemic failures have been identified that severely hinder urban sustainability efforts and civic compliance.

The foremost issue is the extreme cognitive burden placed on everyday citizens at the exact moment of disposal. Without accessible, real-time guidance, households struggle to differentiate between complex packaging materials, often leading to the accidental mixing of highly toxic electronic waste with general biological or solid refuse. This unintentional cross-contamination occurs entirely at the source and cascades through the entire municipal system, ultimately rendering downstream material recovery operations hazardous and economically non-viable.

Compounding this issue is the severe logistical disconnect within municipal collection fleets. City administrators operate with a complete lack of real-time spatial data regarding neighborhood waste density. Consequently, garbage trucks are deployed on static, pre-determined routes that service empty bins while ignoring overflowing zones. This lack of dynamic route optimization results in massive, unnecessary fuel consumption, inflated operational budgets, and an increased urban carbon footprint.

Furthermore, there is a fundamental absence of civic incentivization and workflow transparency. Current frameworks rely solely on passive compliance, offering no gamified or economic rewards to encourage households to invest time in proper segregation. Additionally, there is no streamlined tracking mechanism connecting citizens, dispatchers, and field collectors. Without a centralized, data-driven Business-to-Government (B2G) dashboard, authorities possess no mechanism to verify task completion (e.g., via photographic proof), enforce localized policies, or visualize the ground-level realities of their sanitation networks, leaving the entire system disjointed and highly inefficient.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM OVERVIEW

The proposed system, CleanCity, is a comprehensive, AI-powered web application meticulously designed to modernize urban waste management through a multi-tiered digital ecosystem. The platform directly bridges the gap between household waste generation and municipal logistics by providing specialized, interconnected dashboards for citizens, city authorities, and waste collectors.

For citizens, the platform acts as an interactive tool for responsible disposal. The flagship feature is the AI Waste Scanner, which integrates a custom Google Teachable Machine model via TensorFlow.js to process live camera feeds. This edge-computing module instantly classifies scanned objects into Solid, Bio-waste, or E-waste categories directly within the browser, providing immediate segregation guidance. Citizens can easily drop a pin on an interactive map to schedule a pickup, track their assigned collector's approach in real-time, and earn "Green Points" for successfully completed collections to redeem eco-friendly rewards.

For municipal authorities, the system offers an administrative command center to oversee city-wide sanitation. Administrators can manage incoming pickup requests and assign them to specific drivers. The dashboard visualizes real-time request density through an interactive Live Waste Heatmap powered by Leaflet.js. Furthermore, it replaces static collection schedules with a dynamic Route Optimizer that ingests geolocated requests and utilizes the Open Source Routing Machine (OSRM) Trip API to calculate the most mathematically efficient, multi-stop driving paths for the fleet.

For the operational workforce, waste collectors are provided with a streamlined Worker Dashboard. This interface displays their assigned jobs, enables turn-by-turn map navigation to the pickup coordinates, and enforces accountability by requiring drivers to upload photographic proof of the collected waste to formally complete the task.

5.2 EDGE-AI WASTE CLASSIFICATION (TENSORFLOW.JS)

The core innovation for eliminating source-level contamination is the Edge-AI Waste Scanner. Rather than relying on cloud-based APIs that introduce latency and privacy concerns, the system utilizes a custom object detection model trained via Google Teachable Machine. This model is integrated directly into the frontend architecture using TensorFlow.js, allowing the inference to run entirely within the user's mobile browser. When a citizen points their camera at an item, the model instantly analyzes the visual data against its trained classes (e.g., Plastic, Cardboard, Glass, Metal) and maps it to the primary municipal categories: Solid, Bio-waste, or E-waste.

5.3 GEOSPATIAL LOGISTICS AND OSRM PATHFINDING

To resolve the severe inefficiencies of static municipal routing, the system employs advanced geospatial logistics. When a citizen schedules a pickup, their exact GPS coordinates are captured and stored as pending waypoints. The Authority Dashboard then aggregates these points within specific zones. By integrating the Open Source Routing Machine (OSRM) Trip API, the platform algorithmically sequences these unorganized geographic nodes into the most time- and distance-efficient driving path (solving variations of the Traveling Salesperson Problem). The resulting optimized route, complete with turn-by-turn polyline coordinates and estimated durations, is then rendered on an interactive Leaflet.js map.

5.4 CIVIC DATA VISUALIZATION AND BEHAVIORAL GAMIFICATION

The platform utilizes complex data visualization to monitor both civic compliance and logistical performance. The B2G Authority Dashboard employs Chart.js to translate incoming request data into actionable analytics, including 7-day categorical bar charts and 30-day volume trend lines, allowing city planners to track localized waste generation. Furthermore, a Live Waste Heatmap clusters high-density pickup requests, creating a spatial visual gradient (from green to red) that instantly highlights areas requiring immediate sanitation focus. On the user side, this data loop is sustained by Behavioral Gamification. By interacting with the AI Scanner and successfully completing verified pickups, citizens accumulate "Green Points," which are tracked in a dedicated ledger and can theoretically be redeemed for eco-vouchers .

5.5 PROPOSED SYSTEM ARCHITECTURE

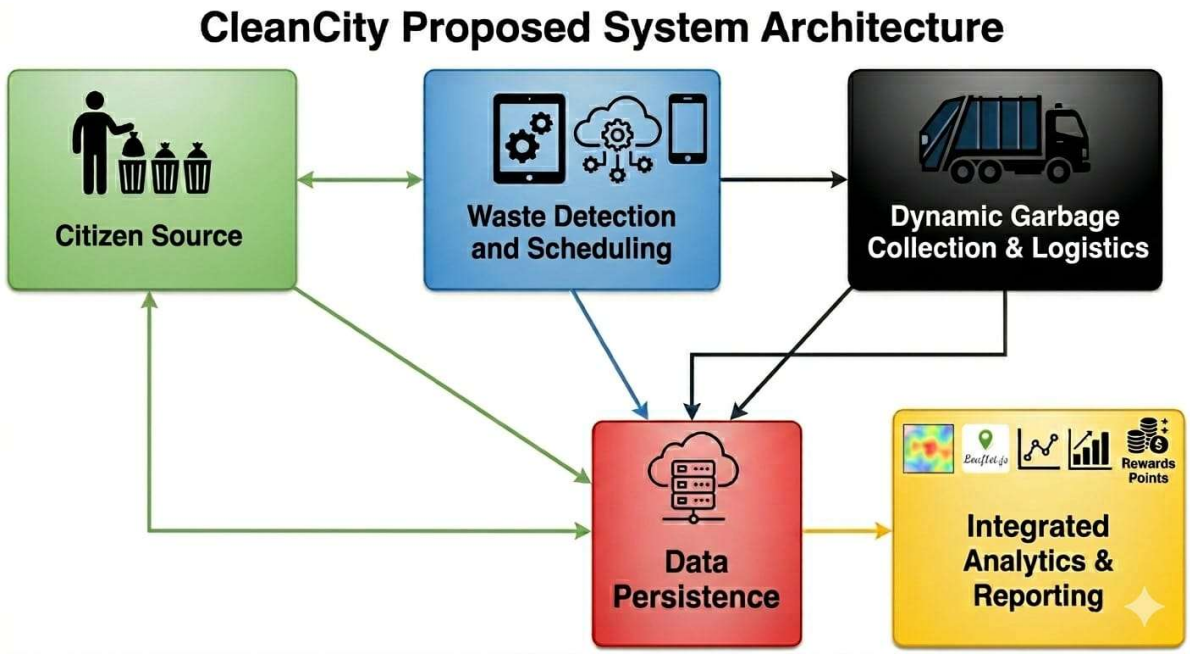

Figure 5.5 Proposed System

The architectural design of the CleanCity ecosystem is partitioned into a multi-tiered structure, ensuring optimal data flow between the public endpoint, the administrative hub, and the field workers.

5.6 ADVANTAGES

- **Zero Source Contamination:** The edge-deployed AI scanner eliminates cognitive friction, guaranteeing highly accurate household segregation before waste ever enters the collection stream.
- **Maximized Fleet Efficiency:** Dynamic OSRM pathfinding directly slashes municipal fuel consumption, vehicle wear-and-tear, and carbon emissions by ensuring trucks only travel optimized routes to verified pickup locations.
- **High Civic Engagement:** The integrated "Green Points" ecosystem transforms a mundane chore into a rewarding habit, fostering a sustainable, community-driven circular economy.
- **Real-Time Administrative Oversight:** The Authority Dashboard provides live heatmaps and comprehensive statistical charts, granting city planners the empirical data necessary to deploy resources strategically.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Citizen Device (Client-Side):

Device Type: Smartphone or Tablet (Android/iOS).

Processor: Quad-core or higher (capable of running TensorFlow.js locally).

RAM: Minimum 2 GB (4 GB recommended for smooth AI inference).

Sensors: Integrated Camera (for AI scanning) and GPS Module (for location tagging).

Internet Connection: Standard 4G/LTE (for initial asset loading and database synchronization).

Collector / Worker Device (Field Operations):

Device Type: Smartphone or Tablet (Android/iOS).

Processor: Standard mobile processor.

RAM: Minimum 2 GB.

Sensors:

Always-On GPS Module: Critical for continuous background location tracking and turn-by-turn navigation (updating every 5 seconds).

Camera: Required to capture and upload photographic proof of completed pickups.

Internet Connection: Continuous 4G/LTE (essential for live map updates and real-time task synchronization).

Authority Device (Administrative-Side):

Device Type: PC / Laptop.

Processor: Intel Core i3 or equivalent.

RAM: Minimum 4 GB (8 GB recommended for complex map rendering and data visualization).

Display: Minimum resolution of 1024x768 (1920x1080 recommended for optimal dashboard viewing).

6.2 SOFTWARE REQUIREMENTS

The software requirements for the CleanCity platform are based on modern, open-source web technologies to ensure cross-platform compatibility and rapid deployment.

Operating System:

Client / Worker: Android, iOS, or any modern mobile OS.

Admin/Development: Windows 10/11, macOS, or Linux.

Frontend Technologies:

HTML5, CSS3 (Custom Neo-Brutalism System), Vanilla JavaScript.

Artificial Intelligence / Machine Learning:

Google Teachable Machine (for model training).

TensorFlow.js (for on-device edge inference).

Mapping and Routing:

Leaflet.js (for interactive map rendering).

OpenStreetMap (for map tiles).

OSRM (Open Source Routing Machine) API (for dynamic route optimization).

Data Visualization:

Chart.js (for generating statistical charts and graphs).

Backend / Database:

Currently: Advanced Browser LocalStorage (for rapid prototyping).

Target Production: Firebase (Auth, Firestore, Hosting).

Development Tools:

Visual Studio Code (or equivalent IDE).

Git & GitHub (for version control).

Python (for running the local development server).

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The CleanCity system is divided into six major modules, each responsible for handling a specific executive functionality within the platform. These modules work together to provide a complete career optimization and data-driven civic auditing system. The system consists of the following main modules:

- Authentication & Security Module
- Citizen Dashboard & Gamification Module
- AI Waste Scanner & Guide Module
- Pickup Scheduling & Logistics Module
- Route Optimization & Tracking Module
- Authority & Collector Operations Module

7.2 MODULE DESCRIPTION

7.2.1 Authentication & Security Module

This module serves as the primary gateway to the platform, managing user identity through a secure registration and login framework. It utilizes asynchronous JavaScript functions to simulate advanced authentication, supporting both standard Email/Password verification and OAuth integration (e.g., Google Sign-In).

A critical feature of this module is its role-based access control (RBAC), which seamlessly categorizes new users and redirects them to their respective specialized interfaces: Citizen, Authority, or Collector. The current architecture relies on secure LocalStorage sessions, establishing a robust foundation primed for migration to a cloud database environment like Firebase.

7.2.2 Citizen Dashboard & Gamification Module

The Citizen Dashboard is the personalized home screen for everyday users, designed to transform mundane waste disposal into an interactive experience. It provides a centralized hub where users can access all critical actions: scanning waste, scheduling pickups, tracking collectors, and consulting the segregation guide.

To ensure sustained civic engagement, this module integrates a comprehensive "Green Points" reward system. Users accumulate points upon the successful verification of their waste pickups.

The dashboard tracks this point history in a dedicated ledger and offers a conceptual redemption marketplace where users can exchange points for eco-friendly rewards, such as partner store discounts or municipality certificates.

7.2.3 AI Waste Scanner & Guide Module

This module represents the core "DeepTech" innovation of the platform, explicitly designed to eliminate source-level contamination. It embeds a custom-trained Google Teachable Machine model directly into the web application using TensorFlow.js.

When a user accesses their camera via the module, the edge-AI instantly processes the live video feed or uploaded image, accurately classifying the item into Municipal Solid, Bio-waste, or E-waste streams. The model provides immediate visual feedback, including a confidence percentage score.

This is complemented by the Segregation Guide, a searchable digital directory containing specific disposal protocols for over 30 common household items, ensuring users have access to definitive recycling knowledge.

7.2.4 Pickup Scheduling & Logistics Module

This module empowers citizens to formalize their waste disposal through precise logistical requests. Utilizing an interactive Leaflet.js map, users can drop a geographical pin or utilize HTML5 Geolocation to tag their exact coordinates.

The system captures the selected waste type (Solid, Bio, or E-waste), the estimated quantity, and the preferred pickup time, compiling this data into a structured request object.

These requests are immediately aggregated into the Authority Dashboard, where city administrators can filter them by status (Pending, Assigned, Completed) and manually or automatically assign them to active waste collectors operating in specific municipal zones.

7.2.5 Route Optimization & Tracking Module

This module fundamentally upgrades municipal collection from static scheduling to dynamic pathfinding. Once authorities aggregate a batch of assigned pickup coordinates, the system leverages the Open Source Routing Machine (OSRM) Trip API to algorithmically sequence the geolocated waypoints.

It solves complex routing problems to output the most mathematically efficient, multi-stop driving path, rendering the turn-by-turn polyline directly onto a master map alongside total distance and time estimates.

Concurrently, the module facilitates live fleet tracking. Citizens can monitor the real-time GPS location of their assigned garbage truck approaching their residence, updating dynamically to provide accurate ETAs and eliminate missed collections.

7.2.6 Authority & Collector Operations Module

This module dictates the backend administration and field execution of the sanitation network.

The B2G Authority Dashboard provides city planners with a centralized command center featuring Live Waste Heatmaps (Leaflet.js) to visualize neighborhood request densities and graphical analytics (Chart.js) to monitor fleet completion rates over time.

For field execution, waste collectors utilize a mobile-optimized Worker Dashboard. This interface displays their assigned optimal routes, allows them to initiate turn-by-turn navigation via external map links, and critically, mandates the capture and upload of photographic proof of the collected waste to formally close the task loop in the database.

CHAPTER 8
SYSTEM TESTING

8.1 UNIT TESTING

Unit testing involves testing individual components or logic blocks of the CleanCity ecosystem independently to ensure that each unit performs as expected before being integrated into the larger workflow. In this system, unit testing is performed on the following core components:

Authentication Validation: Testing the loginWithEmail and loginWithGoogle functions within auth.js to ensure secure role-based assignment and session persistence.

Database CRUD Operations: Validating the LocalStorage wrapper functions in database.js, such as createPickupRequest, assignCollector, and completeRequest, to ensure accurate data formatting and retrieval.

AI Inference Logic: Testing the classifyImage function in classifier.js by passing test image tensors to the Google Teachable Machine model and verifying that it accurately returns the correct rawClass, category, and confidence parameters.

Map Marker Generation: Verifying that specific mapping utilities, like addPickupMarker and addCollectorMarker in maps.js, correctly instantiate Leaflet nodes with the appropriate custom HTML and CSS styling.

8.2 INTEGRATION TESTING

Integration testing evaluates how different modules of the system communicate and transfer data, particularly focusing on the flow between asynchronous APIs and the Document Object Model (DOM). In this system, integration testing includes:

Scanner-to-DOM Integration: Ensuring that the output from the TensorFlow.js model dynamically updates the result-badge, confidence-fill bar, and result-points fields on the scanner.html interface without requiring a page reload.

Geolocation-to-Database Sync: Testing the integration between the HTML5 Geolocation API on the schedule.html page and the database.js backend, ensuring that exact latitude and longitude coordinates.

8.3 SYSTEM TESTING

System testing involves testing the complete CleanCity platform as a single, cohesive unit to ensure it meets all high-level functional requirements across different user roles. This phase tests the full end-to-end operational loop:

End-to-End Workflow: A citizen logs in, uses the AI Scanner to classify waste, and drops a pin on the map to schedule a pickup. The system verifies that this request immediately appears on the Authority Dashboard's heatmap. The authority assigns a collector, and the system verifies that the status updates to "ASSIGNED". The collector logs into the Worker Dashboard, navigates to the location, and uploads a photo proof to complete the job. Finally, the system confirms that the citizen receives the appropriate "Green Points" in their rewards ledger.

Role-Based Security: Ensuring that a Citizen cannot access Authority analytics or Route Optimizer pages, and that unauthorized sessions are automatically redirected to the login index.

8.4 PERFORMANCE TESTING

Performance testing is conducted to evaluate the platform's responsiveness, particularly regarding edge-computation and API latency. In this system, performance testing involves:

Edge AI Inference Speed: Evaluating the latency of the classifyImage function when loading the Teachable Machine model.json and metadata.json files, ensuring that the visual scan executes rapidly on mobile processors without causing the browser to freeze.

Routing Latency: Testing the response time of the getOptimizedTrip function when submitting large arrays of waypoints to the external OSRM API, ensuring the routing polyline renders quickly on the Authority Dashboard.

Map Rendering: Verifying that the Leaflet.js canvases (e.g., heatmap, route-map, track-map) initialize smoothly and handle the dynamic clustering of multiple heatMarkers without significant frame drops.

8.5 USABILITY TESTING

Usability testing focuses on evaluating the user experience, accessibility, and intuitive nature of the dual-sided interface. In this system, usability testing involves:

Mobile Responsiveness: Ensuring that the .grid-2, .grid-3, and .grid-4 CSS layouts properly collapse into a single-column stack on screens narrower than 768px, making the dashboards fully functional on standard smartphones.

Aesthetic Clarity (Neo-Brutalism): Verifying that the high-contrast color palette (--green, --yellow, --blue, --red) paired with bold borders and strict drop-shadows provides clear visual hierarchies and legible text, especially outdoors where waste collectors operate.

Navigation Fluidity: Checking the ease of switching between core tabs (Scan, Schedule, Track, Rewards) via the mobile toggle menu (.nav-toggle), ensuring an intuitive flow for everyday citizens.

CHAPTER 9
RESULT AND DISCUSSION

The CleanCity ecosystem was successfully designed, developed, and tested as a unified, AI-driven platform for modernizing urban waste management. The system effectively bridges the gap between household disposal habits and macro-level municipal logistics by consolidating edge-AI scanning, dynamic geographic pathfinding, and behavioral gamification into a single web application.

The implementation produced the following critical outcomes:

Successful Edge-AI Integration: The platform successfully embedded a custom Google Teachable Machine model via TensorFlow.js. This enabled the AI Waste Scanner to instantly classify loose household refuse into Solid, Bio-waste, and E-waste categories entirely within the user's mobile browser.

Dynamic Logistical Optimization: The Authority Dashboard successfully integrated the Open Source Routing Machine (OSRM) API. The system ingested unorganized, geolocated citizen pickup requests and dynamically sequenced them into mathematically optimized, turn-by-turn collection routes, bypassing the inefficiencies of static municipal scheduling.

Comprehensive Data Visualization: The B2G administrative interface successfully utilized Leaflet.js to generate Live Waste Heatmaps that accurately clustered request densities. Simultaneously, Chart.js modules provided robust analytics regarding waste volume trends and collector completion rates.

Verified Operational Loop: The Worker Dashboard successfully facilitated field operations, allowing waste collectors to view optimized routes and formally close the task loop by uploading photographic proof of collection.

Civic Gamification: The integration of the "Green Points" reward system successfully established a behavioral incentive framework.

In conclusion, CleanCity successfully addresses the major limitations of existing sanitation frameworks by providing a unified, data-driven solution prioritizing Edge Intelligence and Dynamic Logistics. By empowering citizens with real-time segregation tools and providing authorities with actionable geospatial data, the system lays a scalable, software-first foundation for a sustainable circular economy.

CHAPTER 10
CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The CleanCity project successfully demonstrates the development of a high-performance, AI-driven software ecosystem that empowers citizens and modernizes municipal waste management. By consolidating edge-based computer vision for waste scanning, interactive geospatial scheduling, OSRM-powered route optimization, and behavioral gamification into a single, unified "Smart City" platform, the system effectively addresses the severe logistical fragmentation found in traditional urban sanitation.

A pivotal achievement of the platform is its utilization of edge intelligence. The successful integration of a Google Teachable Machine model via TensorFlow.js proves that accurate, real-time waste classification can be achieved natively on consumer smartphones. This effectively eliminates source-level contamination by providing everyday users with instant, offline-capable segregation guidance, completely bypassing the high costs and privacy concerns associated with server-side AI processing.

Furthermore, the implementation of dynamic, mathematically optimized routing via the Authority Dashboard signifies a major leap forward from static fleet management. By providing city planners with live density heatmaps and automated logistical pathfinding, the framework drastically reduces fuel consumption and ensures timely collections. The concurrent integration of a mobile Worker Dashboard ensures total operational transparency, verifying task completion through photographic proof and closing the data loop.

Overall, CleanCity bridges the gap between individual civic habits and macro-level administrative operations. It proves that a scalable, software-first approach—leveraging the existing mobile device infrastructure of the public—can successfully foster a sustainable circular economy, making it an essential blueprint for modernizing urban sanitation networks globally.

10.2 FUTURE ENHANCEMENTS

Although the current CleanCity platform provides a powerful foundation for smart waste management, several strategic enhancements can be implemented to further expand its functionality and impact:

Expanded Localized AI Datasets: The Google Teachable Machine model can be continuously trained on increasingly diverse, localized datasets (e.g., highly specific regional food packaging or crushed materials) to push the visual classification accuracy beyond its current capabilities and handle highly ambiguous waste streams.

Predictive AI Waste Generation Models: Future iterations could implement machine learning algorithms on the Authority Dashboard to analyze historical pickup data, predicting seasonal or event-based surges in waste generation for specific neighborhoods before they occur.

Integration with Smart Bin Sensors: While designed as a software-first solution, the ecosystem's backend could be expanded via APIs to ingest data from physical ultrasonic fill-level sensors installed in high-traffic public bins, merging citizen requests with automated hardware alerts on the Live Heatmap.

Dynamic Third-Party Rewards API: The "Green Points" system could be integrated with live E-commerce or municipal billing APIs, allowing citizens to instantly redeem their earned points for direct property tax rebates, utility bill deductions, or verified discounts at partnered eco-friendly businesses, creating a true, frictionless circular economy loop.

APPENDIX – A
SOURCE CODE

A.1 Authentication and Role Management (auth.js)

```
async function loginWithEmail(email, password) {
  try {
    const user = {
      id: 'user_' + Math.random().toString(36).substr(2, 9),
      email: email,
      user_metadata: { full_name: email.split('@')[0] }
    };
    localStorage.setItem('currentUser', JSON.stringify(user));
    return { success: true, user: user, isNew: false };
  } catch (e) {
    return { success: false, error: e.message };  }
}
async function logout() {
  try {
    localStorage.removeItem('currentUser');
    localStorage.removeItem('pendingAuthRole');
    return true;
  } catch (e) {
    console.error('Logout error:', e);
    return false;  }
}
function requireAuth(callback) {
  const isInSubfolder = window.location.pathname.includes('/citizen/') ||
window.location.pathname.includes('/authority/');
  const loginUrl = isInSubfolder ? '../index.html' : 'index.html';
  const userJson = localStorage.getItem('currentUser');
  if (!userJson) {
```

```
    window.location.href = loginUrl;
  } else {
    if (callback) callback(JSON.parse(userJson));
  }
}
```

A.2 Database Helpers (database.js)

```
function getStorage(key, defaultVal) {
  const data = localStorage.getItem(key);
  return data ? JSON.parse(data) : (defaultVal || []);
}
function setStorage(key, val) {
  localStorage.setItem(key, JSON.stringify(val));
}
async function createPickupRequest(userId, wasteType, quantity, lat, lng, locationName,
scheduledTime) {
  const requests = getStorage('pickupRequests');
  const newRequest = {
    id: 'req_' + Math.random().toString(36).substr(2, 9),
    userId: userId,
    wasteType: wasteType,
    quantity: quantity,
    lat: lat,
    lng: lng,
    locationName: locationName || 'Unknown location',
    status: 'pending',
    scheduledTime: new Date(scheduledTime).toISOString(),
    createdAt: new Date().toISOString()
  };
  requests.push(newRequest);
  setStorage('pickupRequests', requests);
  return newRequest.id;
}
```

```
async function completeRequest(requestId, userId, wasteType) {
  const pointsMap = { 'Solid': 15, 'Bio-waste': 20, 'E-waste': 25 };
  const points = pointsMap[wasteType] || 10;
  const requests = getStorage('pickupRequests');
  const idx = requests.findIndex(r => r.id === requestId);
  if (idx !== -1) {
    requests[idx].status = 'completed';
    requests[idx].completedAt = new Date().toISOString();
    setStorage('pickupRequests', requests);
  }
  const users = getStorage('users', {});
  if (users[userId]) {
    users[userId].greenPoints = (users[userId].greenPoints || 0) + points;
    setStorage('users', users);  }
}
```

A.3 AI Waste Classification (classifier.js)

```
Const  MODEL_URL =
'https://teachablemachine.withgoogle.com/models/YOUR_MODEL_ID/' ;
const CLASS_MAP = {
  'Cardboard': { category: 'Bio-waste',    points: 20, badge: 'badge-bio'    },
  'Glass':     { category: 'Solid Waste',  points: 15, badge: 'badge-solid'  },
  'Trash':     { category: 'E-waste',      points: 25, badge: 'badge-ewaste' }
};
let classifierModel = null;
let isModelLoaded = false;
async function loadModel() {
  try {
    const modelURL    = MODEL_URL + 'model.json';
    const metadataURL = MODEL_URL + 'metadata.json';
    classifierModel = await tmImage.load(modelURL, metadataURL);
    isModelLoaded = true;
    return classifierModel;
```

```
  } catch (e) {
    console.warn('Could not load model:', e.message);
    isModelLoaded = false;
    return null;  }
}
async function classifyImage(model, imageElement) {
  if (model && isModelLoaded) {
    try {
      const predictions = await model.predict(imageElement);
      const top = predictions.reduce(function(a, b) {
        return a.probability > b.probability ? a : b;
      });
      const mapped = CLASS_MAP[top.className] || { category: 'Unknown', points: 0,
badge: 'badge-solid' };
      return {
        rawClass: top.className,
        confidence: Math.round(top.probability * 100),
        category: mapped.category,
        points: mapped.points,
        badge: mapped.badge      };
    } catch (e) {
      console.error('Classification error:', e);    }
  }}
```

A.4 Map and OSRM Routing Utilities (maps.js)

```
function initMap(containerId, lat, lng, zoom) {
  var map = L.map(containerId).setView([lat, lng], zoom || 13);
  L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
    attribution: '&copy; OpenStreetMap contributors',
    maxZoom: 19
  }).addTo(map);
  return map;
}
```

```
async function getOptimizedTrip(waypointsArray) {
  if (!waypointsArray || waypointsArray.length < 2) {
    return { orderedWaypoints: waypointsArray || [], routeCoordinates: [] };
  }
  var coordString = waypointsArray.map(function(wp) {
    return wp.lng + ',' + wp.lat;
  }).join(';');
  var url = 'https://router.project-osrm.org/trip/v1/driving/' + coordString +
    '?roundtrip=true&source=first&destination=last&geometries=geojson';
  try {
    var response = await fetch(url);
    var data = await response.json();
    if (data.code === 'Ok' && data.trips && data.trips.length > 0) {
      var trip = data.trips[0];
      var routeCoords = trip.geometry.coordinates.map(function(c) {
        return [c[1], c[0]];      });
      var orderedWaypoints = data.waypoints
        .sort(function(a, b) { return a.waypoint_index - b.waypoint_index; })
        .map(function(wp) {
          return    {    lat:    wp.location[1],    lng:    wp.location[0],    waypointIndex:
wp.waypoint_index };
        });
      return {
        orderedWaypoints: orderedWaypoints,
        routeCoordinates: routeCoords,
        distance: trip.distance,
        duration: trip.duration      };
    }
  } catch (e) {
    console.error('OSRM trip error:', e);
  }
}
```

APPENDIX – B

SCREENSHOTS

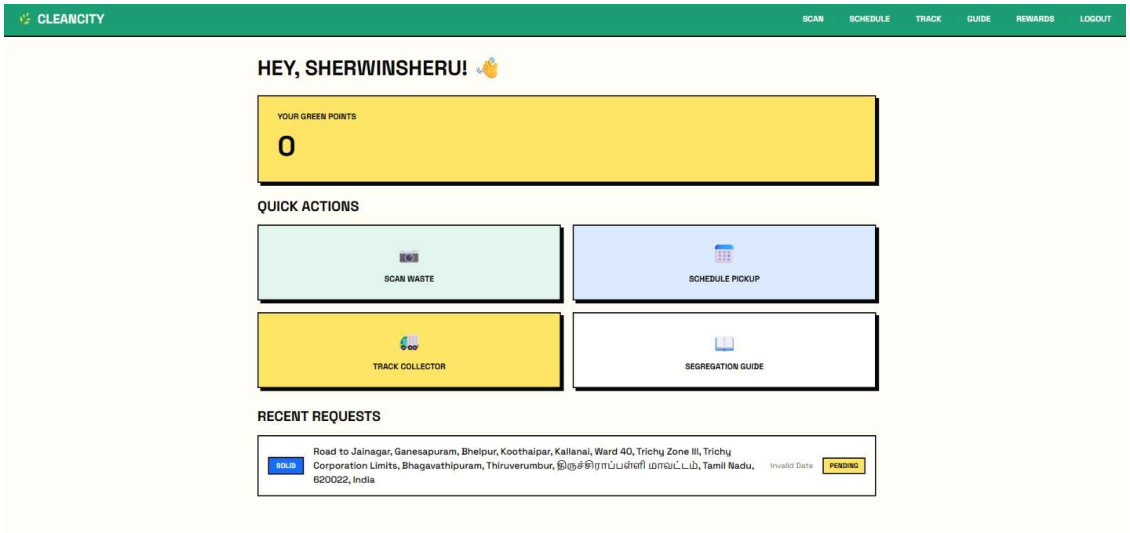

Figure B.1 Citizen Dashboard and Green Points Ledger

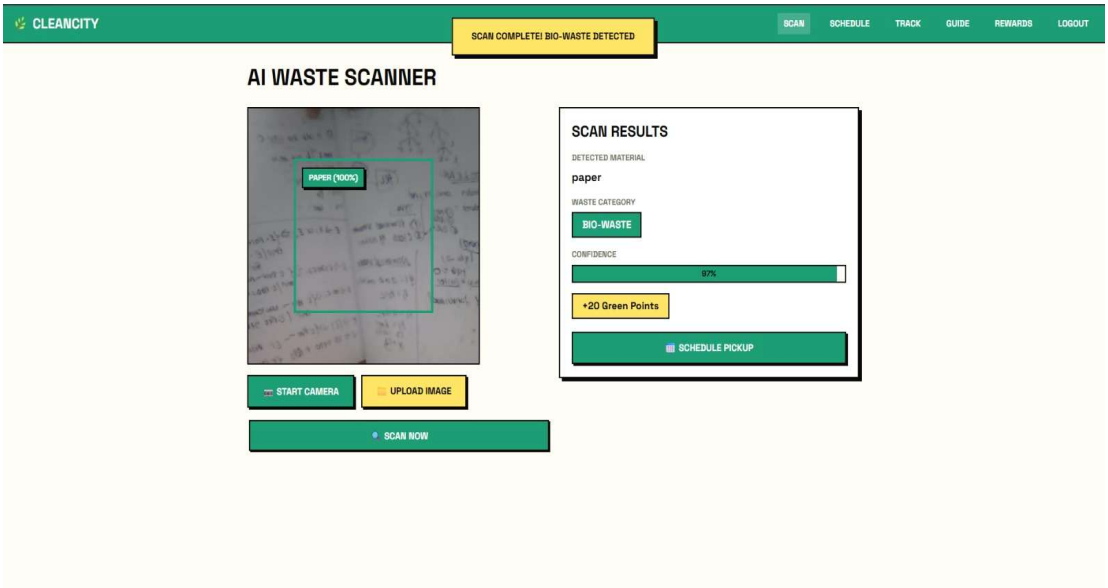

Figure B.2 AI Waste Scanner Real-Time Classification

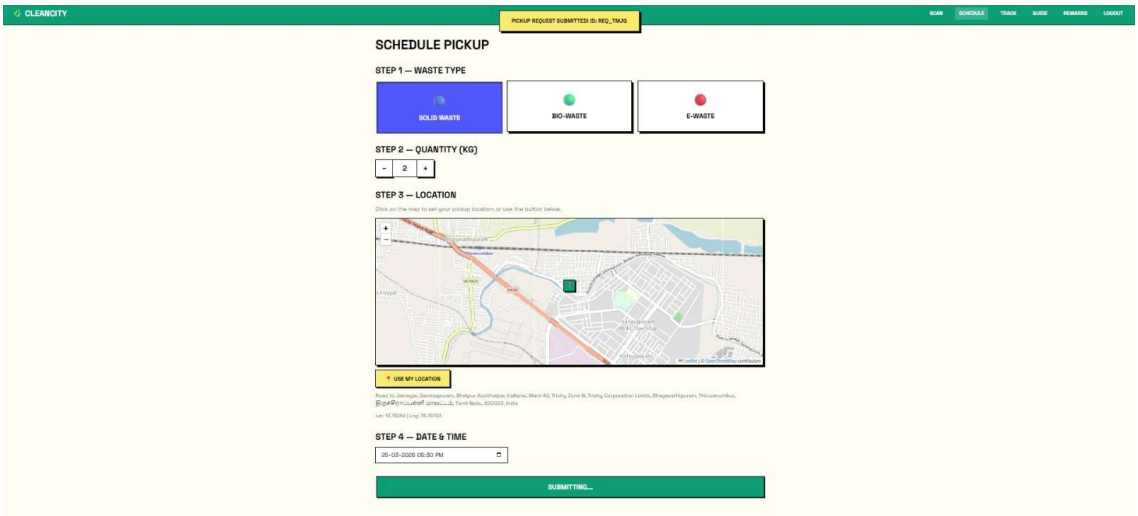

Figure B.3 Geospatial Pickup Scheduling Interface

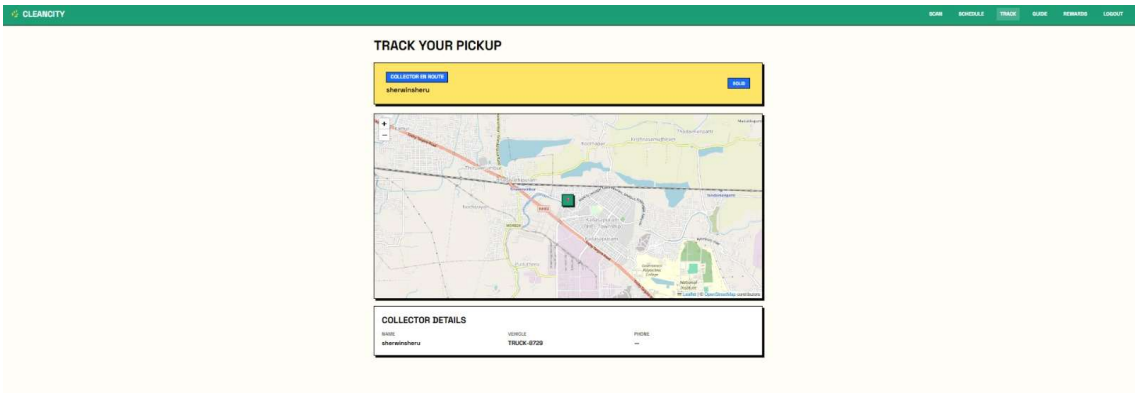

Figure B.4 Citizen Live Tracking Map

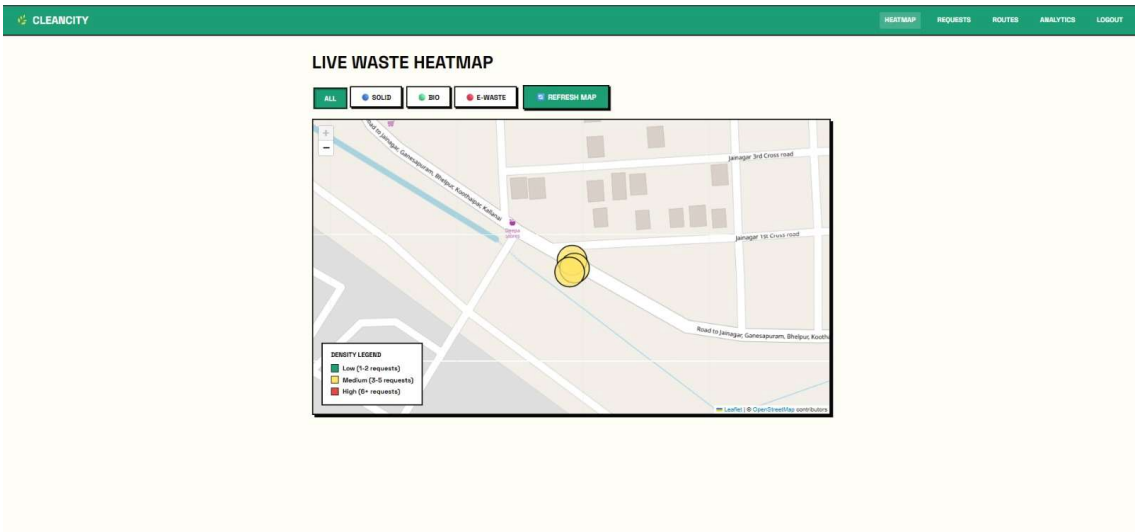

Figure B.5 Live Waste Heatmap Visualization

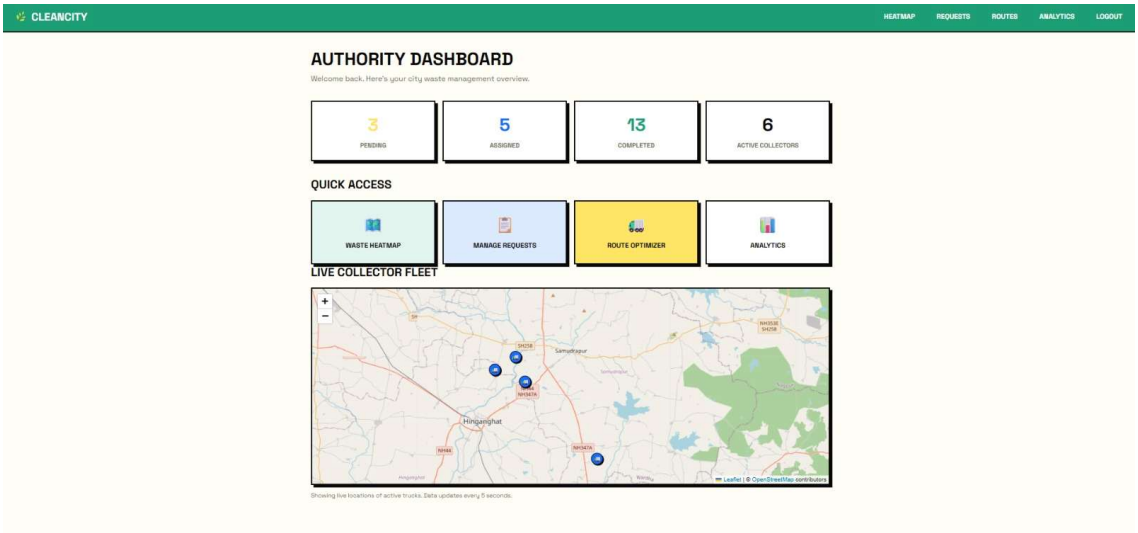

Figure B.6 Authority Dashboard

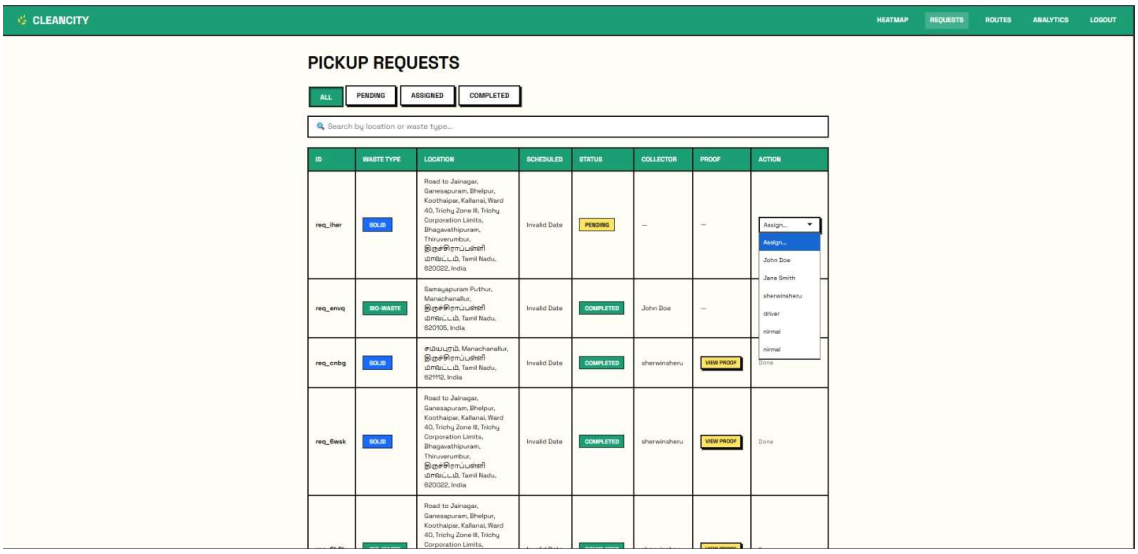

Figure B.7 Authority Request Management Table

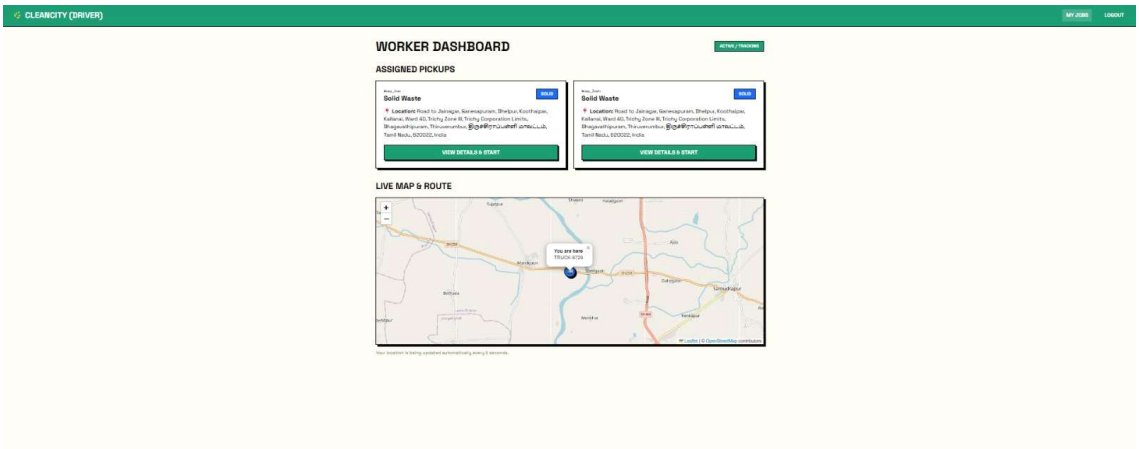

Figure B.8 Collector/Worker Task List Interface

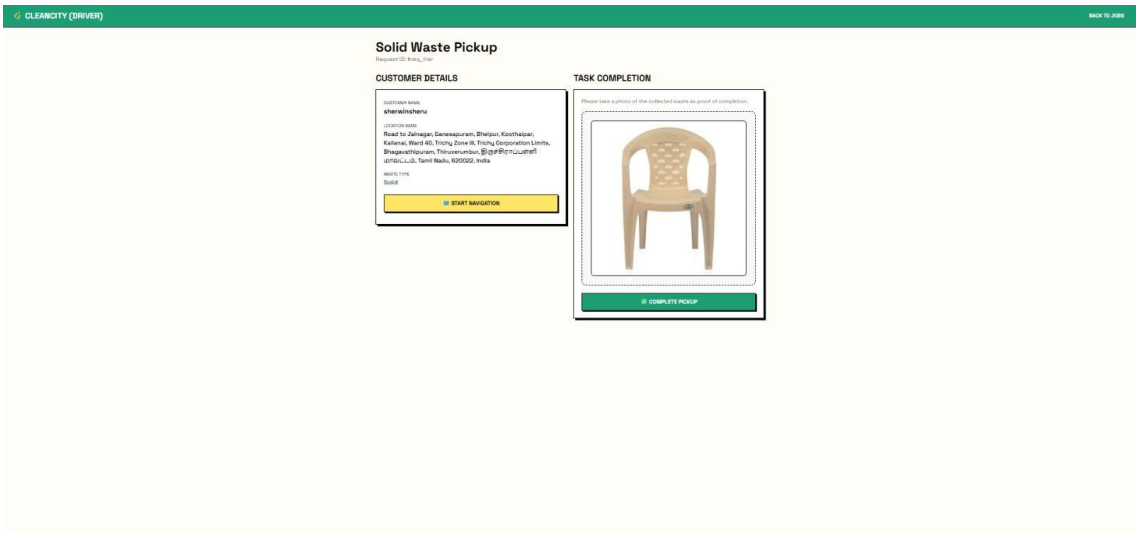

Figure B.9 Photographic Proof Upload Interface

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